

**TED(15)- 4134
(REVISION-2015)**

**MODEL QUESTION PAPER (REV-15)
FOURTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY**

OPERATING SYSTEMS

(Maximum Marks : 100)

TIME : 3HRS

PART - A

Answer all Questions in one or two sentences. Each question carries **2 marks**.

I.

1. List the functions of a Loader.
2. Define starvation.
3. Name four CPU scheduling algorithms.
4. List two address binding schemes
5. Define virtualization

5x2=10

PART - B

Answer any **FIVE** of the following questions. Each question carries **6 marks**.

II.

1. Compare multiprogramming and multiprocessing.
2. Explain PCB with its structure.
3. Explain deadlock avoidance.
4. Explain first-fit, best fit allocation strategies.
5. Compare logical and physical addresses.
6. Explain six file operations.
7. List the limitations of virtualization.

5x6=30

PART - C

Answer **One** full question from each Unit. Each question carries 15 marks.

UNIT I

III.

- a. Explain Network management and file management components of OS **9**
- b. Explain real time systems **6**

OR

IV.

- a. Explain assembler, compiler and interpreter **9**
- b. Explain loader and its functions **6**

UNIT II

V.

- a. Explain Multilevel queue and Multilevel Feedback Queue 9
- b. Explain resource allocation graph with a diagram 6

OR

VI.

- a. Explain long term, short term and medium term schedulers with diagram 9
- b. Explain SJF scheduling and find the average waiting time for the following processes.

Process	Burst time	
P1	6	
P2	8	
P3	7	
P4	3	6

UNIT III

VII .

- a. Explain Demand paging with diagram 9
- b. Explain the Virtual memory concepts 6

OR

VIII.

- a. Explain FIFO ,optimal and LRU page replacement algorithms 9
- b. List the steps to handle page fault with diagram 6

UNIT IV

IX.

- a. Compare contiguous, linked and indexed file allocation methods 9
- b. Explain the need for virtualization 6

OR

X.

- a. Explain memory, storage and data virtualizations 9
- b. Distinguish between Sequential and Indexed file organizations 6

15x4=60